

MIXED STRATEGY NE

23/10/25

$$A_i = \left\{ \begin{matrix} p & 1-p \\ C, & NC \end{matrix} \right\} \quad p \text{ is probability} \quad p \in [0, 1]$$

* Players are allowed to randomise over their choices.

For mixed strategy NE, the probability distribution has to be optimal.
 ↘ Stochastic steady state

Eg:-

	H	T	
H	1, -1	-1, 1	p
T	-1, 1	1, -1	1-p
	q	1-q	

MSNE

$$p = \frac{1}{2}, \quad q = \frac{1}{2}$$

Ans. Let $(\frac{1}{2}, \frac{1}{2})$ are the probabilities of H & T deployed by 2 and $(p, 1-p)$ is deployed by 1.

$$\begin{aligned} P(1 \text{ gets 1 Re}) &= P(HH \cap TT) = P(HH) + P(TT) \\ &= \frac{p}{2} + \frac{1-p}{2} \\ &= \frac{1}{2} \end{aligned}$$

$$P(1 \text{ loses } -1) = \frac{1}{2}$$

Since $P(1 \text{ gets } 1Rc) = P(1 \text{ loses } 1Rc)$ every p is optimal
 $\Downarrow$
 $p = \frac{1}{2}$ is optimal

Similarly from 2's POV

$p = \frac{1}{2}, q = \frac{1}{2}$ is optimal

Eg :-

		Probability	Distribution	
		l_1	l_2	
a	200	0.5	0	l - lottery
b	100	0	1	
c	2	0.5	0	

Lotteries

$l_1 > l_2$ - Risk taking behaviour

$l_2 > l_1$ - Risk averse

Expected Utility Theory

people are allowed to randomise

If players have Von-Neumann Morgenstern (VN-M) preference then their payoff from a lottery $(p_1, p_2 \dots p_n)$ over a set of n outcomes $(x_1, x_2 \dots x_n)$ is $V_j(p_1, p_2 \dots p_n, x_1, \dots, x_n) = \sum_{i=1}^n p_i u_j(x_i)$
 $u_j(x_i)$ are called Bernoulli functions.

Eg:-

	b_1	b_2	
a_1	pq	$p(1-q)$	p
a_2	$(1-p)q$	$(1-p)(1-q)$	$1-p$
	q	$1-q$	

$$\alpha_2 = (q, 1-q)$$

$$\begin{aligned} V_1(p, 1-p, q, 1-q) &= p \left(q u_1(a_1, b_1) + (1-q) u_1(a_1, b_2) \right) \\ &\quad + \\ &\quad (1-p) \left(q u_1(a_2, b_1) + (1-q) u_1(a_2, b_2) \right) \\ &= p E_1(a_1, \alpha_2) + (1-p) E(a_2, \alpha_2) \end{aligned}$$

Here, payoffs are not just for ordinal purposes, the values matter. Next game illustrates this difference.

STAG HUNT

	St	H
St	2,2	0,1
H	1,0	1,1

(A)

	St	H
St	4,4	0,1
H	1,0	1,1

(B)

Claim: These games are not the same if they are Vn-M.

l_1	l_2	
0.5	0	St, St
0.5	0	St, H
0	1	H, St
0	0	H, H

For P1, $l_1 \sim l_2$ in A

$$U_1(l_1) = \frac{1}{2} \times 2 + \frac{1}{2} \times 0 + 0(1+1) = 1$$

$$U_1(l_2) = 1 \times 1 = 1$$

Now, in game B

$$U_1(l_1) = \frac{1}{2} \times 4 + \frac{1}{2} \times 0 = 2$$

$$U_1(l_2) = 1 \times 1 = 1$$

For A, $l_1 > l_2$ in game B

Mixed Strategy Equilibrium

A mixed strategy of a player i is a probability distribution over her action set.

α_i : Mixed strategy of i

$$\alpha_i(a_k) = p, \quad 0 < p \leq 1 \quad a_k \in A_i$$

$$\sum_{a_k \in A_i} \alpha_i(a_k) = 1 \quad ; \quad \text{Eg: } \alpha_i = (0.4, 0.6)$$

Mixed strategy profile: $(\alpha_1, \alpha_2, \dots, \alpha_n) = \alpha$

Definition of MSNE

α^* is an MSNE iff $\forall i \quad u_i(\alpha^*) \geq u_i(\alpha_i, \alpha_{-i}^*)$ for each α_i of player i .

Eg:- Matching Pennies

	H	T	
H	1, -1	-1, 1	$\left. \begin{matrix} p \\ 1-p \end{matrix} \right\} \alpha_1$
T	-1, 1	1, -1	
	$\underbrace{\begin{matrix} q & 1-q \end{matrix}}_{\alpha_2}$		

$$u_1(\alpha) = p(E_1(H, \alpha_2)) + (1-p) E_1(T, \alpha_2)$$

$$= p(2q-1) + (1-p)(1-2q)$$

If $2q-1 > 1-2q$ then optimal $p=1$

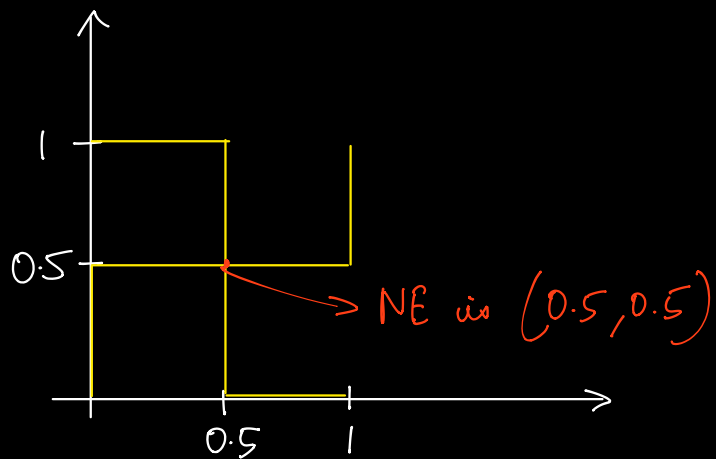
$$\downarrow$$

$$q > \frac{1}{2}$$

If $q < \frac{1}{2}$ then optimal $p=0$

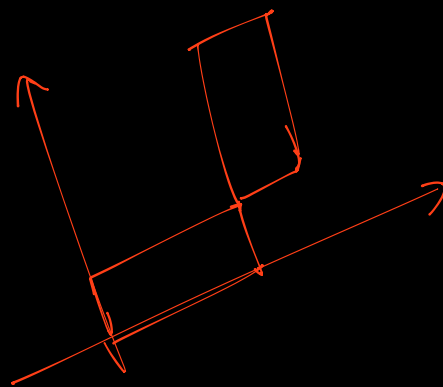
If $q = \frac{1}{2}$ then optimal $p = \frac{1}{2}$

Necessary for plotting BSR



Note: Although this game does not have an NE, it has an MSNE.

Joy BOS with
mixed strategy



$(0, 0) \quad (1, 1)$
Pure strategy
NE

$(\frac{2}{3}, \frac{1}{3})$
proper MSNE

$$v_i(\alpha) = \sum_{a_i \in A_i} \alpha_i(a_i) E_i(\alpha_i, a_i)$$

weighted
average

α^* is an MSNE iff for each i the expected payoff to every action to which α_i^* assigns +ve probability is the same (given α_{-i}^*).

And expected payoffs of actions assigned zero probability is at most the expected payoff of actions assigned +ve probability.

Logic: On the contrary if E of all actions were not the same, player i could play the action with the highest E for sure.

If it were higher than the rest, the player could have played that action for sure

Q.

	x	y	z
a	6, 5	2, 4	9, 4
b	2, 6	10, 10	10, 50
c	16, 2	1, 0	4, 10

PT $\alpha_1 = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right)$

$\alpha_2 = \left(\frac{1}{3}, 0, \frac{2}{3}\right)$ is an MSNE

Ans. For 1 $E_1(a, \alpha_2) = 6 \times \frac{1}{3} + 0 + 9 \times \frac{2}{3} = 8$

$E_1(b, \alpha_2) = 8$

$E_1(c, \alpha_2) = 8$

For 2 $E_2(\alpha_1, x) = 5 \times \frac{1}{2} + 6 \times \frac{1}{4} + 2 \times \frac{1}{4} = 4.5$

$E_2(\alpha_1, y) = 4 \times \frac{1}{2} + 10 \times \frac{1}{4} = 4.5 \leq 4.5$

$E_2(\alpha_1, z) = 4.5$

$\therefore$ it is an MSNE

MIXED STRATEGY DOMINATION

Eg :-

	L	R
T	1	1
M	4	0
B	0	3

No dominant action for player 1
if mixed strategy is not used.

Claim: $(0, p, 1-p)$ strictly dominates T for what p ?

$$U_1(\alpha_1, L) = 4p > U_1(T, L) = 1 \quad \text{--- (1)}$$

$$U_1(\alpha_1, R) = 3(1-p) > U_1(T, R) = 1 \quad \text{--- (2)}$$

$$\text{From (1) and (2)} \quad p \in \left(\frac{1}{4}, \frac{2}{3}\right)$$

Strict domination

Player i 's mixed strategy α_i strictly dominates her action a_i iff

$$U_i(\alpha_i, a_{-i}) > U_i(a_i, a_{-i}) \quad \forall a_{-i}$$

These actions are
pure strategy ones and
not mixed

WEAK DOMINATION

Player i 's mixed strategy α_i strictly dominates her action a_i iff

$$u_i(\alpha_i, a_{-i}) \geq u_i(a_i, a_{-i}) \quad \forall a_{-i}$$

and there is at least one a_{-i} s.t.,

$$u_i(\alpha_i, a_{-i}) > u_i(a_i, a_{-i})$$

Recall: A player never plays a str dominated action in an NE. Same holds for MSNE.

Q.

	a	b	c
x	2, 3	1, 1.5	4, 1
y	4, 0	3, 2	2, 4

Find all NE.

means both NE and MSNE.

$\alpha_2 = (p, 0, 1-p)$ str dominates b.

$$u_1(x, \alpha_2) > u_1(x, b) \Rightarrow 3p + 1(1-p) > 1.5$$
$$\Rightarrow p > \frac{1}{4}$$

$$u_1(y, \alpha_2) > u_1(y, b)$$



$$4(1-p) > 2$$

$$p < \frac{1}{2}$$

$$\Rightarrow \frac{1}{4} < p < \frac{1}{2}$$

EXTENSIVE GAMES IN PERFECT INFORMATION

Eg:- Chess

1. Set of players

2. Set of actions is dependent on the current state.

↓
Instead we use terminal histories (h)

3. Player function, $P(h)$
↙ ↘
not terminal history gives the player who moves at each point in the terminal history
↘
complete sequence of actions from beginning to end after which no more actions can be taken

4. Preferences of a player is defined over terminal histories

Eg:- ENTRY GAME

① Players

In incumbent - Person holding power currently

Challenger - Person contemplating entry into the market.

Sequence - Challenger plays first, IN or OUT

② List of terminal histories

① (OUT)

② (IN, Accommodate)

③ (IN, Fight)

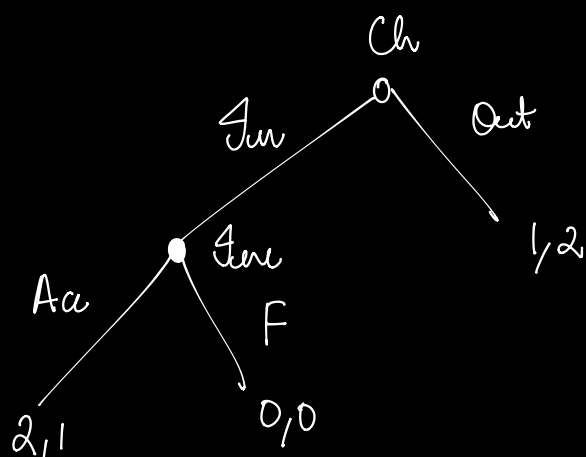
$$P(\phi) = Ch, \quad P(IN) = True$$

③ Preferences

Payoffs	Ch	True
2	IN, Acc	OUT
1	OUT	IN, Acc
0	IN, F	IN, F

} Preferences between terminal histories

④ Game Tree



Discussion of NE

① (IN, Acc) ✓

② (OUT, Acc) ✗

③ (OUT, F) ✓

→ Player 1 could profitably deviate to IN

$(a^1, a^2 \dots a^k) \rightarrow \text{history}$

Subhistory: $(a^1, a^2 \dots a^m) \quad m \in [1, k] \text{ and } \phi$

* Suppose, h is a non-terminal history, $P(h) = i$, say
Let (h, a) is a history then $A_i(h) = \{a : (h, a) \text{ is a history}\}$

A STRATEGY $\rightarrow$ is a set of actions

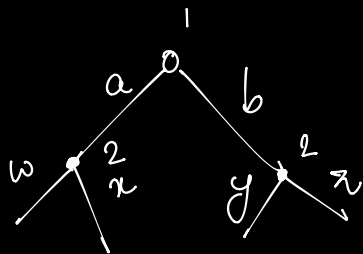
For player i , s_i is a strategy

For any h (a non-terminal history) such that $P(h) = i$, s_i specifies an action from A_i .

Eg:- $P(h) = \text{Ch}$ for $h = \phi$, $S_{\text{Ch}} = \{\text{IN, OUT}\}$

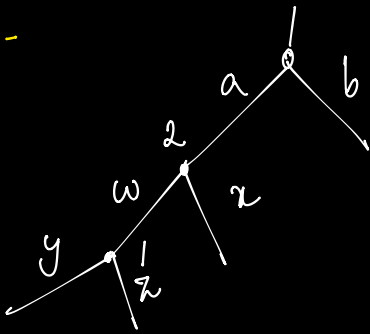
$P(h) = \text{Jue}$ for $h = \text{IN}$, $S_{\text{Jue}} = \{\text{F, Acc}\}$

Eg:-



A strategy profile generates an outcome $O(s)$.

Eg:-



$$S_1 = \{ay, az, by, bz\}$$

$$S_2 = \{w, z\}$$

$$s = (az, w) \rightarrow O(a, w, z)$$

Outcomes are
always terminal
histories

SEQUENTIAL PROCEDURE

A strategy profile s^* is an NE if for each i

$$u_i(o(s^*)) \geq u_i(o(s_i, s_{-i}^*)) \text{ for all } s_i \in S_i$$

$S_1, S_2, \dots, S_n$
 $\downarrow \quad \downarrow \quad \downarrow$
 $A_1, A_2, \dots, A_n$
 $s_i \in S_i$

Preferences:

Preferences from the action profiles are the same as preferences from strategy profile.

Eg:- ENTRY GAME

In the strategic form of the extensive game:

Players: Ch, Inc

Actions: $A_{Ch} = \{IN, OUT\}$, $A_{Inc} = \{F, Acc\}$

Preferences given by the following payoff matrix

	F	Acc
IN	0,0	2,1 → NE
OUT	1,2	1,2

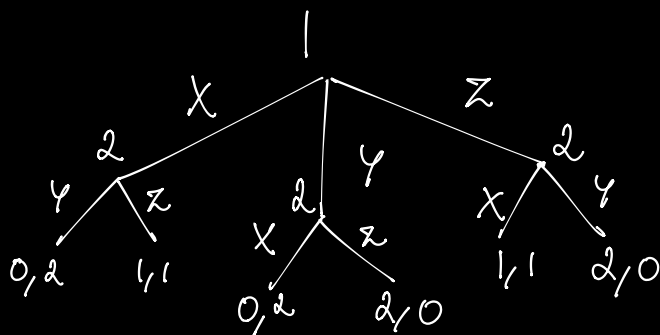
Eg:- mj-draft, Page 186

3 policies X, Y, Z

2 players 1 and 2

1	2
X	Z
Y	Y
Z	X

Ans. Game Tree



Players: Player 1 and 2

Terminal histories : $(X, A) : Y, Z$
 $(Y, B) : X, Z$
 $(Z, C) : Y, Z$

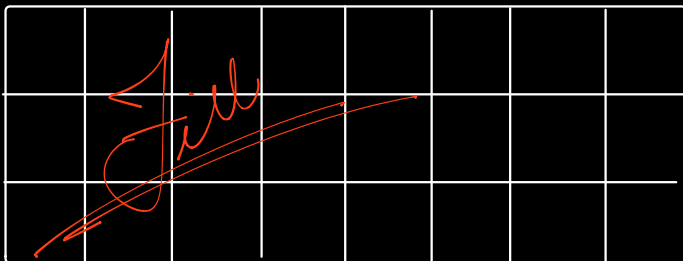
Preferences

$$u_1(x, y) = u_1(y, x) = u_2(z, y) = u_2(y, z) = 0$$

Strategies

$$S_1 = \{x, y, z\} \longrightarrow A_1$$

$$S_2 = \left\{ \begin{array}{l} yxx, yxy, yzx, yzy, \\ zxx, zxz, zzy, zzz \end{array} \right\} \longrightarrow A_2$$



Claim: NE does not make sense in extensive games.

↓
Eg:- NE: (Out, F) does not make sense in the entry game.
↳ Difficult to interpret as a steady state in an extensive game.

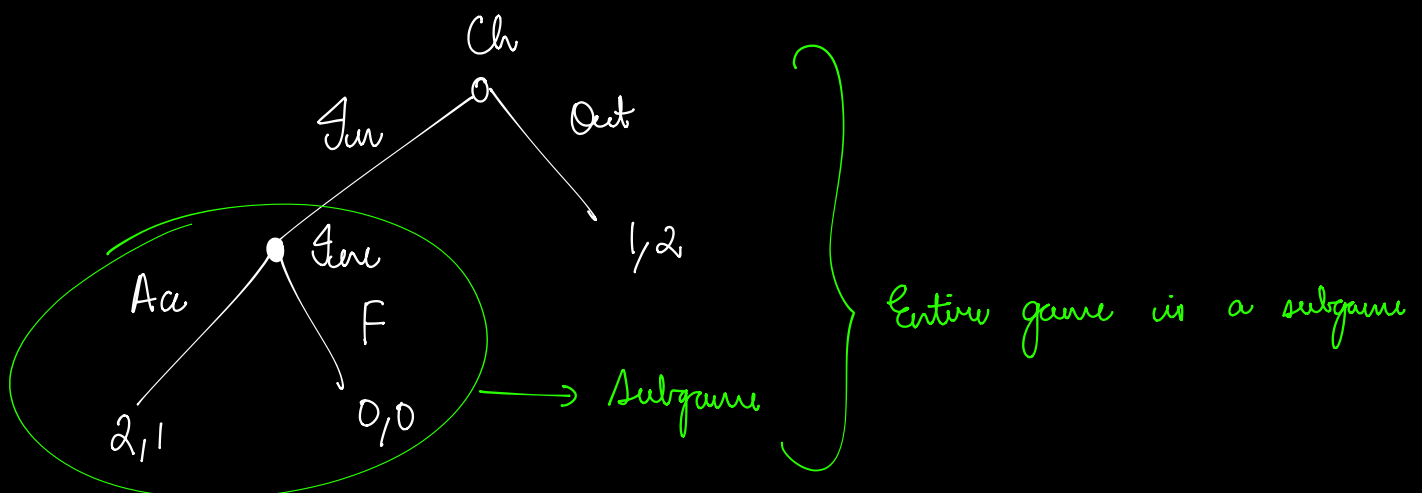
(Out, F) is not a perturbed steady state

SUBGAME PERFECT EQ (SPE/SPNE)

Subgame of a game: For any non-terminal history of a game Γ , the part of the game that follows after h , $\Gamma(h)$ is a subgame.

Eg:- ENTRY GAME

Non-terminal history = ϕ, IN



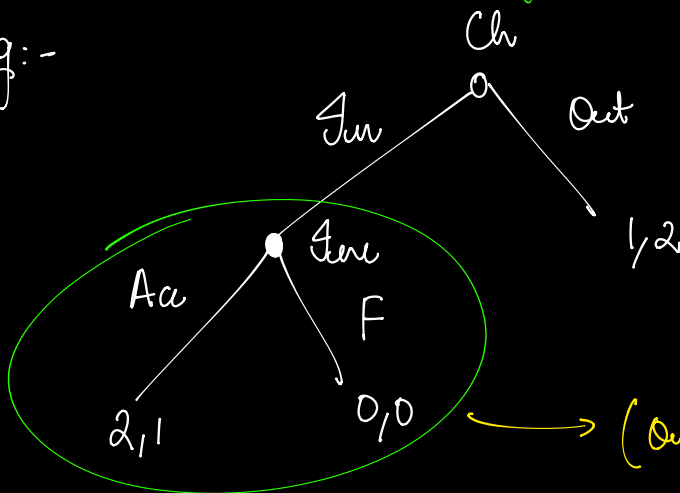
An **SPE** introduces **NE** in every subgame of the original game.

For a game Γ , s^* is an SPE if for every h ST $P(h) = i$,

$$u_i(o_h(s^*)) \geq u_i(o_h(s_i, s_{-i}^*))$$

For every h , in every game, h could be IN or ϕ

Eg:-



(Out, F) is not an NE in this subgame

Process that rules out (Out, F) as an SPE.

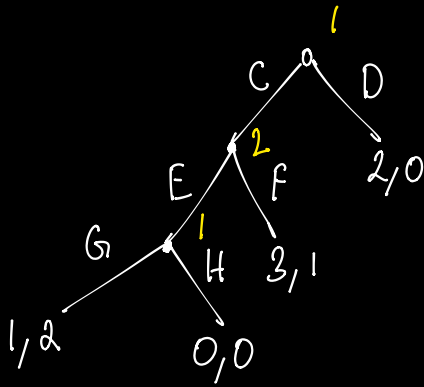
Take $h = IN$, $P(h) = In$

$$o_h(s^*) = (IN, F) \implies u_i(o_h(s^*)) = 0$$

$$o_h(s_{-i}, s_{-i}^*) = (IN, Acc) \implies u_i(o_h(s_{-i}, s_{-i}^*)) = 1$$

SPE: Perturbed steady state

II nd game



NE: $(CH, F), (DG, E), (DH, E)$

Only SPE

EASIER METHOD TO FIND SPE (Backward Induction)

* Only if the game has a finite horizon

I METHOD (Above tree is used as example)

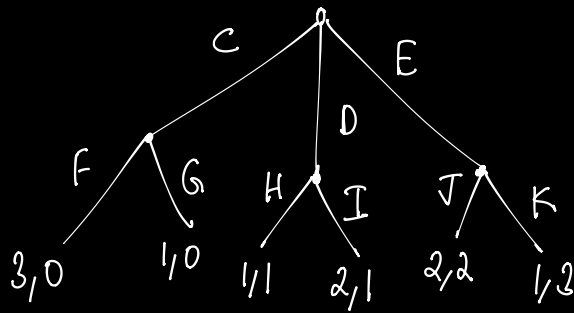
1. Take subgames of length: 1
2. Optimal action of player 1 $\Rightarrow \underline{G}$

- ### II
1. Take subgames of length: 2
 2. Optimal action of player 2 given player 1 would play $G \Rightarrow \underline{E}$

- ### III
1. Take subgames of length: 3
 2. Optimal action of player 1 given actions E and G would be played $\Rightarrow \underline{D}$

Strategy: (DG, E) is the SPE

Eg:-



I

$\Gamma(C) : P(C) = 2$, optimal action : F & G

$\Gamma(D) : P(D) = 2$, " : H & I

$\Gamma(E) : P(E) = 2$, " : K

Optimal combinations of actions (strategies) : FHK, FIK, GHK, GIK

II

Worst	FHK,	optimal action of	$I = C$
" "	FIK,	" "	$I = C$
" "	GHK,	" "	$I = C, D, E$
"	—— Incomplete ——		

